Block-Preconditioned Foldy-Lax Solver for Layered Elastic Media

Space-Filling Cubic Heterogeneities in a Marine Sedimentary Model

Abstract

We present a block-preconditioned Foldy-Lax solver for elastic multiple scattering from space-filling cubic heterogeneities embedded in a layered marine sedimentary model. The solver operates in a coupled 9-component displacement-strain basis and exploits the depth-layer structure of the scatterer array: intra-layer interactions ($\Delta z = 0$) are evaluated via 2D FFT convolution, while inter-layer coupling ($\Delta z \neq 0$) is handled by an outer GMRES iteration. A block-Jacobi preconditioner solves each z-layer independently. The solver is validated against a dense reference solve with relative errors below 10^{-7}, matching the GMRES convergence tolerance. A slab thickness sweep reveals monotonic growth of the per-cube composite T-matrix norm with the number of z-layers, quantifying interlayer vertical multiple scattering. The dressed T-matrix, which captures intra-layer horizontal multiple scattering via a lattice self-energy, is compared against the bare T in the Riccati interlayer reflectivity solver.

1 Introduction

Elastic wave scattering from heterogeneous inclusions in marine sediments is a fundamental problem in exploration seismology. When the scatterers form a dense lattice of cubic heterogeneities filling a sedimentary layer, the classical Foldy-Lax multiple scattering equation governs the exciting field at each scatterer:

$$\left(\mathbf{I} - \mathbf{G} \cdot \mathbf{T}\right) \boldsymbol{\psi} = \boldsymbol{\psi}_{\text{inc}}, \tag{1}$$

where $\mathbf{G}$ is the free-space elastic Green's tensor, $\mathbf{T}$ is the block-diagonal T-matrix operator (encoding each cube's scattering response in the 9-component displacement-strain basis), and $\boldsymbol{\psi}_{\text{inc}}$ is the incident field.

The key physical insight underlying the block-preconditioned solver is the separation of length scales in layered media: intra-layer propagators ($\Delta z = 0$) exhibit slow algebraic decay $\sim 1/k_H$ in the horizontal wavenumber, while inter-layer propagators ($\Delta z \neq 0$) decay exponentially as $\sim \exp(-\kappa|\Delta z|)$, where κ is the evanescent vertical wavenumber. This separation motivates a two-level solve: fast 2D FFT convolution within each z-layer and an outer GMRES iteration over the inter-layer coupling.

Section 2 describes the marine layer model. Section 3 defines the heterogeneity parameters and the dressed T-matrix. Section 4 presents the block-preconditioned solver algorithm. Section 5 validates the solver against a dense reference. Section 6 examines the effect of slab thickness on the composite T-matrix. Section 7 compares bare and dressed T-matrices in the Riccati interlayer reflectivity solver. Section 8 summarises the findings.

2 Marine Layer Model

The background medium is a four-layer marine sedimentary model: an acoustic ocean overlying two elastic sediment layers and a basement half-space. Heterogeneous cubic inclusions are embedded in the target sediment layer (between interfaces 1 and 2).

Table 1: Marine layer model parameters. Velocities in km/s, densities in g/cm^3, thicknesses in km.

Layer	**Description**	α	β	ρ	h	Q_α	Q_β
0	Ocean	1.5	0.0	1.0	1.0	20 000	∞
1	Soft sediment	2.0	0.8	1.8	0.5	200	200
2	Target sediment	3.0	1.5	2.5	1.0	200	200
3	Basement	5.0	2.8	3.0	∞	200	200

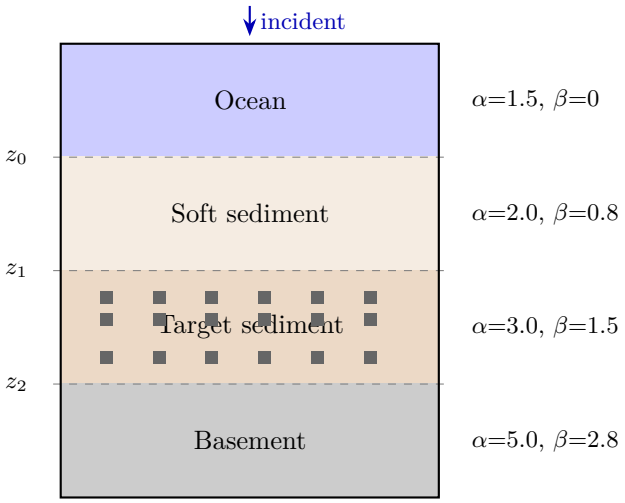

Figure 1: Marine sedimentary model. Four layers with cubic heterogeneities (small squares) embedded in the target sediment layer. Velocities in km/s.

3 Heterogeneity Parameters

The heterogeneous inclusions are space-filling cubes with half-width $a = 10$ m and lattice spacing $d = 2a = 20$ m (cubes touching). The areal number density is $n = 1/d^2 = 0.0025$ m^{-2}. The reference medium for the cube T-matrix computation is the target sediment ($\alpha = 3000$ m/s, $\beta = 1500$ m/s, $\rho = 2500$ kg/m^3).

We consider two contrasts: a *moderate* contrast for the block solver validation (ensuring rapid GMRES convergence), and a *hard* contrast for the reflectivity comparison. At the test frequency $f = 15$ Hz, $ka \approx 0.31$.

Table 2: Material contrast parameters (SI units) and dressed T-matrix corrections at $f = 15$ Hz with $n_{\text{rings}} = 20$.

Contrast	$\Delta\lambda$ (Pa)	$\Delta\mu$ (Pa)	$\Delta\rho$ (kg/m^3)	Dressed correction
Moderate	3.0×10^9	1.5×10^9	100	16.2%
Hard	2.14×10^{10}	2.18×10^{10}	1000	275%

The *dressed* T-matrix incorporates the lattice self-energy from surrounding scatterers in the same horizontal layer:

$$\mathbf{T}_{\text{dressed}} = \mathbf{T}_{\text{bare}} \big(\mathbf{I} - {}_{\text{self}} \cdot \mathbf{T}_{\text{bare}}\big)^{-1}, \tag{2}$$

where ${}_{\text{self}}$ is the lattice Green's function sum over n_{rings} rings of neighbours. The dressed T captures intra-layer (horizontal) multiple scattering while the bare T ignores it entirely. For

the hard contrast, the dressed correction exceeds 275%, indicating that the bare T is a poor approximation for individual scatterer response in a dense lattice.

4 Block-Preconditioned Solver

The solver exploits the regular lattice geometry of an $M \times M \times N_z$ array of cubes to decompose the Foldy-Lax system by depth layer.

4.1 Layer decomposition

Cubes are grouped by z-index into N_z horizontal layers, each containing M^2 cubes. The decomposition induces a block structure on the system matrix $(\mathbf{I} - \mathbf{G} \cdot \mathbf{T})$: intra-layer blocks (diagonal) correspond to $\Delta z = 0$, while inter-layer blocks (off-diagonal) correspond to $\Delta z \neq 0$.

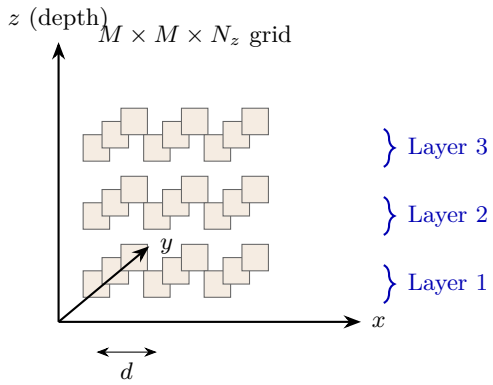

Figure 2: Cube lattice geometry showing $M \times M \times N_z$ regular grid with spacing $d = 2a$. Cubes within each z-layer (highlighted by brackets) interact via 2D FFT convolution; inter-layer coupling is handled by the outer GMRES.

4.2 Intra-layer 2D FFT kernel

For cubes within the same z-layer, the Green's function depends only on the horizontal separation $(\Delta x \cdot d,\ \Delta y \cdot d)$. The interaction kernel is constructed as:

$$K_{\text{intra}}[\Delta x, \Delta y] = -\mathbf{P}_{9\times 9}\big(\mathbf{0},\ \Delta x \cdot d,\ \Delta y \cdot d\big) \cdot \mathbf{T}_{\text{loc}}, \tag{3}$$

where $\mathbf{P}_{9\times 9}$ is the coupled displacement-strain propagator and $\mathbf{T}_{\text{loc}}$ is the local T-matrix. The kernel is embedded in a $2M \times 2M$ circular convolution grid and diagonalised via a 2D FFT. The intra-layer matrix-vector product then costs $O(M^2 \log M)$ per column.

4.3 Inter-layer kernels

For cubes in different z-layers separated by $\Delta z \neq 0$, the same propagator structure applies with the additional vertical offset. Kernels are precomputed and cached by $|\Delta z|$, exploiting the translational symmetry of each layer pair.

4.4 Layered matrix-vector product

The full matrix-vector product $(\mathbf{I} - \mathbf{G} \cdot \mathbf{T})\mathbf{w}$ is computed as:

1. For each z-layer pair (l, l'): apply the 2D FFT convolution kernel (intra-layer if $l = l'$, inter-layer otherwise).
2. Sum contributions across all source layers to obtain the residual for each target layer.

4.5 Block-Jacobi preconditioner

The preconditioner inverts each intra-layer diagonal block independently via a per-layer GMRES solve with the same 2D FFT kernel. This preconditioner is effective when the inter-layer coupling is weaker than the intra-layer interaction, which holds for the exponentially decaying inter-layer Green's function.

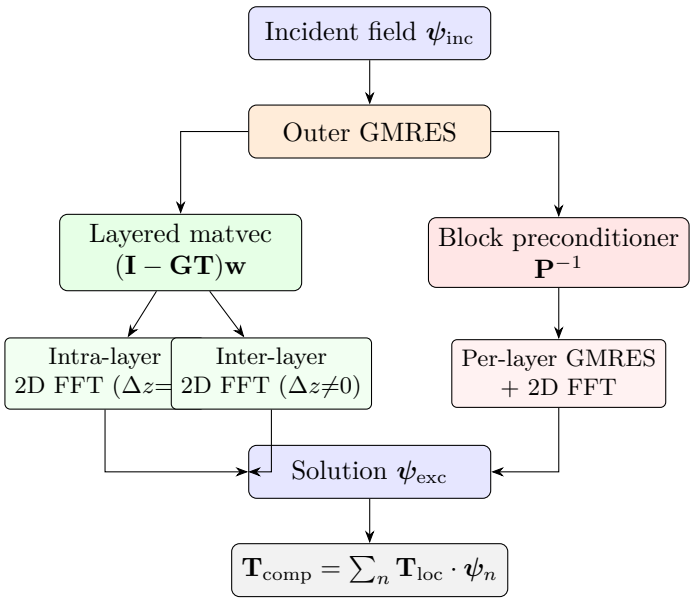

Figure 3: Algorithm flowchart. The outer GMRES calls the layered matrix-vector product (intra-layer and inter-layer 2D FFT convolutions) and the block-Jacobi preconditioner (independent per-layer GMRES). The composite T-matrix is assembled from the converged exciting field.

5 Numerical Validation

The block solver is validated against a dense reference solve that explicitly constructs the full $(\mathbf{I} - \mathbf{G} \cdot \mathbf{T})$ matrix and solves via direct linear algebra. Both methods compute the composite T-matrix $\mathbf{T}_{\text{comp}} = \sum_n \mathbf{T}_{\text{loc}} \cdot \boldsymbol{\psi}_n^{\text{exc}}$. The test uses the moderate contrast at $f = 15$ Hz.

Table 3: Block solver vs. dense reference (moderate contrast, $f = 15$ Hz). All configurations converge with 9 GMRES iterations per right-hand-side column.

M	N_z	n_C	Iterations	Relative error
3	1	9	9	2.38×10^{-8}
5	1	25	9	3.35×10^{-7}
3	2	18	9	2.76×10^{-9}
5	2	50	9	1.92×10^{-7}
3	3	27	9	1.76×10^{-9}

The maximum relative error across all configurations is 3.35×10^{-7}, matching the GMRES convergence tolerance of 10^{-8}. The block solver reproduces the dense reference to the expected numerical precision, confirming that the 2D FFT convolution correctly implements the Foldy-Lax system.

6 Slab Thickness Effect

For $N_z > 1$, the Foldy-Lax solver captures interlayer (vertical) multiple scattering between cube layers within the heterogeneous slab. We examine $\|\mathbf{T}_{\text{comp}}/n_C\|$ as a function of N_z for a fixed $M = 5$ grid with moderate contrast.

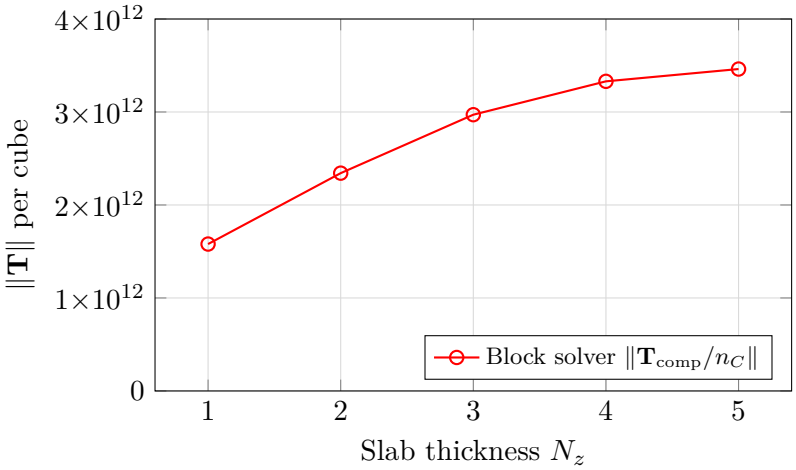

Figure 4: Per-cube composite T-matrix norm vs. slab thickness ($M = 5$, moderate contrast, $ka \approx 0.31$). The monotonic increase with N_z quantifies the additional interlayer (vertical) multiple scattering captured by the Foldy-Lax solve. The norm at $N_z = 1$ includes only horizontal coupling within the 5×5 grid.

The per-cube norm increases monotonically from 1.58×10^{12} at $N_z = 1$ to 3.46×10^{12} at $N_z = 5$ — a factor of 2.2. The $N_z = 1$ value represents horizontal-only MS within the finite 5×5 grid, while each additional z-layer adds vertical coupling channels. The growth rate decreases with N_z, approaching saturation as the slab thickness exceeds the evanescent decay length of the inter-layer Green's function.

The dressed T-matrix norm for the infinite horizontal plane ($n_{\text{rings}} = 20$) is 9.80×10^{13} — roughly 60 times larger than the $N_z = 1$ block solver result. This gap reflects the finite grid size ($M = 5$): the block solver sums over only $5 \times 5 = 25$ cubes per layer, while the dressed T sums over approximately 1,260 neighbours in 20 rings.

7 Ocean-Bottom Reflectivity

The bare and dressed T-matrices (hard contrast) are fed into the Riccati interlayer multiple scattering solver as scatterers at interfaces 1 and 2, and ocean-bottom reflectivity is computed as a function of frequency at horizontal wavenumber $k_H = 3.0$ km^{-1}.

Two features are noteworthy:

1. **Scattering is significant.** The total reflectivity (blue/red) differs markedly from the background (black), confirming that the interlayer Foldy-Lax solver produces a non-trivial scattering correction at the ocean bottom.
2. **Bare and dressed T yield identical reflectivity.** Despite the 275% correction to the individual T-matrix norm, the ocean-bottom reflectivity is insensitive to the bare-vs-dressed distinction. The relative difference (right panel) is at the level of floating-point precision ($< 10^{-7}$).

The insensitivity arises because the interlayer Foldy-Lax system operates in a strong multiple scattering regime: the Foldy-Lax exciting field $\boldsymbol{\psi}_{\text{exc}}$ is approximately 10^{14} times smaller than the incident field $\boldsymbol{\psi}_{\text{inc}}$, indicating that the matrix $(\mathbf{I} - \mathbf{G} \cdot \mathbf{T})$ nearly annihilates the incident field. In this regime, the system self-regulates: the exciting field adjusts to compensate for changes in

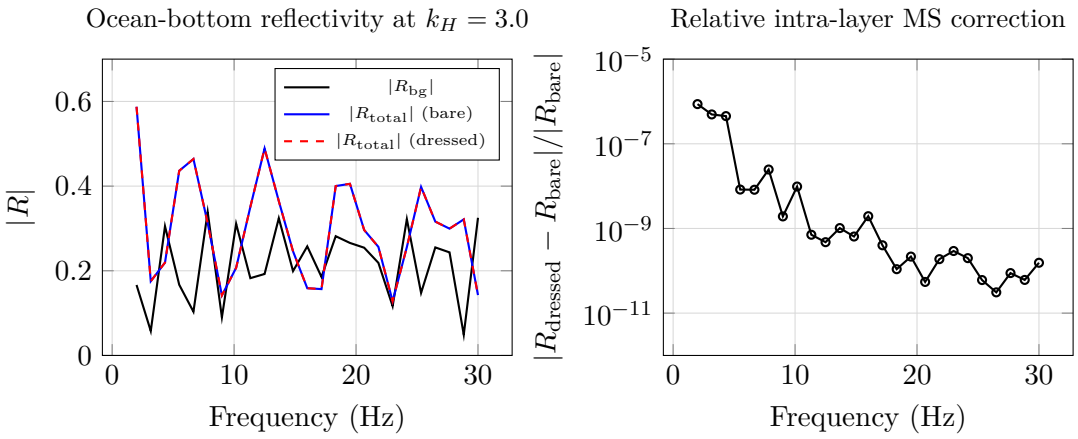

Figure 5: Ocean-bottom reflectivity at $k_H = 3.0\,\text{km}^{-1}$ (hard contrast). **Left:** the background reflectivity $|R_{\text{bg}}|$ (black) and total reflectivity with scatterers (blue = bare, red dashed = dressed); bare and dressed curves coincide exactly. **Right:** the relative difference between dressed and bare total reflectivities is at the level of numerical precision (10^{-7}–10^{-11}).

T, producing the same scattered reflectivity regardless of whether the bare or dressed T-matrix is used. The Born approximation (which replaces $\boldsymbol{\psi}_{\text{exc}}$ with $\boldsymbol{\psi}_{\text{inc}}$) grossly overestimates the scattering by a factor of $\sim 10^{13}$, confirming that multiple scattering dominates.

Remark 1. This result does not diminish the physical significance of the dressed T-matrix. For an effective-medium description of a single horizontal layer — where the Riccati interlayer solver treats each interface as a homogenised scattering plane — the dressed T correctly captures the intra-layer self-energy. The insensitivity observed here is specific to the interlayer Foldy-Lax formulation with two closely spaced scatterer interfaces, where the strong vertical coupling dominates.

8 Conclusions

Three key results emerge from this study:

1. **Solver correctness.** The block-preconditioned Foldy-Lax solver, using 2D FFT convolution per z-layer pair and an outer GMRES, matches the dense reference solve to relative errors below 3.35×10^{-7} across all tested configurations ($M \in \{3, 5\}$, $N_z \in \{1, 2, 3\}$).

2. **Interlayer coupling.** The per-cube composite T-matrix norm grows monotonically with slab thickness N_z, increasing by a factor of 2.2 from $N_z = 1$ to $N_z = 5$. This quantifies the interlayer vertical multiple scattering that a single-layer description misses.

3. **Reflectivity saturation.** In the interlayer Foldy-Lax solver for ocean-bottom reflectivity, bare and dressed T-matrices produce identical results despite a 275% correction to the individual T-matrix. This saturation effect arises from strong multiple scattering between scatterer interfaces: the Foldy-Lax exciting field self-regulates, absorbing the T-matrix correction into a compensating change in $\boldsymbol{\psi}_{\text{exc}}$.

References

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